

Aditya Prabakaran

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Computing student at Imperial College London, currently Site Reliability Engineer on Apple's ML Platforms team, with three years of production experience across five startups. Happiest at the seam between research and production: compilers, infrastructure, agents, and the unglamorous work of keeping them running.

EDUCATION

Imperial College London, MEng in Computing (AI & ML)

2023–2027 (expected)

On track for First Class · GPA 3.8

Coursework: compilers, OS, ML, distributed systems

Hackathons: 1st place, SwyftGesture (hands-free input)

EXPERIENCE

Apple, Site Reliability Engineer, ML Platforms

2026–present

Built a Kubernetes capacity forensics platform end-to-end (collectors, scanners, delta-query UI) used by **30+** SREs to diagnose EC2 capacity exhaustion, saving up to **\$3M** per AWS availability zone annually.

Built a cloud-agnostic capacity request and reservation management system that replaced ad-hoc Slack coordination with an auditable workflow handling hundreds of requests monthly.

Created a Kubernetes manifest validation framework detecting misconfigurations at deploy time; retrospective analysis shows it would have caught **72%** of deployment-related incidents over the prior year.

Kubernetes · Go · AWS · SRE · Distributed systems · Observability · Linux

8x, Full-stack & AI/ML Developer

Apr 2026–May 2026

Optimized production analytics from **24s to sub-second** via SQL-side aggregation and indexed Postgres RPC rewrites; cut /posts payloads **90%+** (22MB → ~1–2MB).

Simplified messaging architecture, deleting ~400 lines of legacy API code while enabling a new admin reply UX.

Resolved **3 critical** production vulnerabilities: org takeover, exposed financial Server Actions, and DB search-path injection across **49 functions**.

PostgreSQL · Next.js · TypeScript · Node.js · Security

Canopy Labs, General Engineer

Mar 2026–Apr 2026

Consolidated a **5-VM** voice-routing architecture into a single gateway service, eliminating 2 dedicated sync VMs.

Replaced 3s polling-based cache sync with Redis pub/sub, cutting cache-update latency to **sub-10ms**.

Built atomic quota enforcement in Redis, closing a client-SDK tampering hole in the API.

Added per-key rate limiting (60s sliding window, configurable RPM) directly in the gateway's WebSocket hot path.

Redis · WebSockets · Gateway architecture · Rate limiting · Distributed systems

Vani, Full-stack & AI/ML Developer

2025–2026

Architected, shipped, and deployed the flagship multi-tenant web app on AWS with Docker + Kubernetes.

Cut p95 backend latency **500ms → 120ms**; scaled to **100+** concurrent users.

Stood up CI/CD: release cadence **2 days → 6 hours**, production bugs **–76%**.

Integrated LLM workflows via Model Context Protocol for clinical-admin automation.

Held **99.9%** uptime through staged rollouts and load-balanced workers.

React · TypeScript · Next.js · FastAPI · Redis · AWS · Kubernetes · Docker

Trajex, Machine Learning Developer

2024–2025

Deployed LLaMA 3.2-7B-Instruct in production for a **20%** cost reduction vs OpenAI and **12%** lower inference latency.

Led product design and built the inference backend.

Pitched investors and onboarded K3 Capital Group as a paying client.

LLaMA 3.2 · Python · Inference optimization · Product

Altus Reach, ML Engineer (Contract)

2024

Built a video saliency model in a team of 3, improving prediction accuracy by **19%**.

Shipped Azure-hosted inference pipeline for production traffic.

Full-stack work on company web app (TypeScript / Next.js / React).

Azure AI · Python · TypeScript · Next.js · Computer Vision

SELECTED PROJECTS

- Project Nine** (PyTorch, Isaac Lab, MuJoCo, PPO): Quadruped robot trained in simulation. Rebuilt the sim from the CAD assembly as a 12-DOF hardware model, cutting the steps needed for a stable walking policy by roughly an order of magnitude. 0.62 m/s sustained, 17.8s standing.
- Kit** (Python, YOLO11n, faster-whisper, openWakeWord): Local-first voice assistant with its own perception and reasoning stack, wired into a self-supervised loop that improves the models from live use.
- Istoria** (Python, CLIP): Scores personal-photo novelty and merges near-duplicates. Rebuilt around distance-from-home and message-volume signals after CLIP turned out to measure visual, not situational, novelty.
- WACC Compiler** (Scala, LLVM, Docker): Front-to-back compiler with full LLVM codegen, GitLab CI/CD and a hosted PaaS deployment.
- Sandbox Orchestrator** (FastAPI, AsyncIO): Orchestration platform for ephemeral execution sandboxes, with producer/consumer queues and blast-radius isolation.
- Llama Distillation** (PyTorch, QLoRA): QLoRA fine-tune of Llama 3.2 3B on years of personal message history; 64% of assessed outputs matched ground-truth tone and style.
- SlopFilter** (JavaScript): Browser extension filtering AI-generated content from social feeds, using on-device Naive Bayes over a hand-curated vocabulary.
- SwyftGesture** (Python, MediaPipe, OpenCV): Hands-free mouse and volume control from webcam hand tracking. First place at a hackathon.

SKILLS

- Languages:** TypeScript, Python, C, C++, Rust, Scala, Haskell, Kotlin, JavaScript
- Systems:** Linux, Docker, Kubernetes, CI/CD, AWS, Azure, Vercel, Redis, WebSockets
- AI / ML:** PyTorch, LLama 3.2, QLoRA, PPO / RL, Isaac Lab, MuJoCo, YOLO11n, faster-whisper, MediaPipe, OpenCV, GraphRAG, MCP, Local LLMs
- Web:** Next.js, React, FastAPI, Node.js, Tailwind, MDX
- Reliability:** Observability, Incident response, Load testing, Distributed tracing
- AI Tools:** Claude Code, Codex CLI, Ollama, CLIP, OpenAI API, MCP

LANGUAGES

- English:** Native
- French:** Native
- Tamil:** Native
- Spanish:** Reading & speaking
- Hindi:** Reading & speaking